S.No. : 351 NAI 4302

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43901 Roll No. | | | | | | | | | | |

B. Tech. Examination, 2025-26

(Odd Semester)

ARTIFICIAL INTELLIGENCE IN MECHANICAL ENGINEERING SYSTEMS

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : 8×1=8

   (a) Define artificial intelligence (AI) in the context of mechanical engineering.

   (b) Name two typical mechanical engineering systems (MES) where AI can be applied.

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(c) State the purpose of a sensor in a automated mechanical system.

(d) What are Actuators? Give one example.

(e) What is CAD and how is AI used in it?

(f) Give two example of process control applications in mechanical system.

(g) Give one example/application of AI in thermal engineering.

(h) What is additive manufacturing?

SECTION – B

2. Attempt any two parts of the following : 2×6=12

(a) Explain the definition, types and applications of mechanical engineering systems (MES). How does AI specifically enahance the performance of these systems?

(b) Illustrate the basic elements of an automated system with neat block diagram. Explain the function of each element.

(c) Explain the general structure and components of an expert system. use a flow chart to illustrate how an expert system processes user queries and provides solutions for a mechanical fault diagnosis problem.

(d) Discuss the application of artificial intelligence in thermal engineering. Provide specific examples in area like HVAC optimization or heat exchanger design.

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. 8×5=40

3. (a) Discuss the different types of artificial intelligence relevant to mechanical engineering applications. Provide a mechanical example for each type.

(b) Discuss the role of machine learning (ML) in the maintenance and optimization of mechanical systems, focussing on fault detection and predictive maintenance.

(c) Describe the fundamental difference between machine learning (ML) and export systems.

4. (a) Elaborate the need for analog-digital conversions in modern mechanical control systems. Explain the working of both ADCs and DACs.

(b) Discuss the various types of control systems with neat sketch. Where would an AI based control system typically fit and why?

(c) Describe the archtiecture and functioning of direct digital contact interfaces used for data transmission in automated mechanical setups.

5. (a) Discuss the importance of knowledge sources and structure characterization in the development of a robust expert system for mechanical design optimization.

(b) Explain the application of AI in CAD/CAPP/MRP for manufacturing. How does AI help in achieving more efficient production life?

(c) Analyze the role of robotics and adaptive control in advanced manufacturing process.

6. (a) How does AI facilitate the implementation of adaptive control for process optimization in a mechanical plant?

(b) Describe the specific application of AI in additive manufacturing (3D printing). How is AI used for process monitoring, defect detection and material selection?

(c) Elaborate on the application of AI in 3D design and modeling. How can AI algorithms aid a mechanical engineer in generating innovative and optimized designs?

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